

ActInf Textbook Group ~ Cohort 2 ~ Meeting 4 (Chapter 2, part 1)

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Hello. It is September 23rd, 2022, and we are in meeting four for Cohort 2 of the Active Inference Textbook Group. We're having our first discussion of the Chapter 2 of the textbook.

So, before we jump to the book and assess where things are at with questions, does anyone want to provide any overview, thoughts, or experiences that they had while reading or rereading Chapter 2?

Just overall, anything about Chapter 2?

There's a lot of math.

Okay, Blue, thank you. Anyone else?

Okay.

We have this questions table, and none have added any questions specifically for Chapter 2, or if someone did and put them somewhere else, just let us know.

These are really important. We hope and expect and prefer that people do have questions while they're reading the text.

And this is a really essential learning practice, and it helps improve our epistemic niche.

It also contributes to future cohorts and future learners when we can have, again, every single kind of question.

From the directly related, what does this word mean here, to the tangential, every kind of question that people have, it's a huge service to actually just add several of them to this table.

So we don't have any for today, and that will free us up to give an overview of Chapter 2, walk through it, and kind of connect a few dots, perhaps.

And then it'd be awesome in our second discussion of Chapter 2 for people to have noted down a few things that they were curious about, including during this discussion people can add directly or write down somewhere else and add later.

But this is a really important way to structure our learning individually and collectively.

So we're going to go into Chapter 2 and walk through it, paying attention to the formalisms and also to a lot of the work that has been added through the Math Learning Group and through various people's efforts to improve the legibility of the math, because it'll be like either a first coat of paint or a refresher for truly everyone.

So everyone's going to be approaching this and trying to learn the math in different ways.

Let's start with Figure 1.2.

This is the high road coming from the top and the low road coming from the bottom.

Chapter 2 is going to be leading with the low road.

Chapter 3 is going to be presenting from the high road.

So Chapter 2 is going to begin with an overview of Bayes' theorem, Bayesian statistics, and then make its way up through some of these terms described here and reach that kind of common meeting point with active inference.

Before we flip to Chapter 2, does anyone have any just general comments on the low road?

Chapter 3 is going to be leading with the low road.

Okay.

The chapter begins with this quote from William James.

My thinking is first and last and always for the sake of my doing.

What does anyone think that means or how does it reflect on them?

Why is it here?

Why is it here?

If anyone can just unmute an address or raise their hand.

Sean?

I suppose it's making the connection between perception and action there with the thinking and the doing.

which is important in chapter two

awesome

as the name active inference would have it there are going to be many connections between cognition and inference and action like the embodiment of maneuvers in the real embodied world as well as even ways of thinking about mental or cognitive behaviors as actions like attention as action so that will be a thread that continues as we continue to build on active inference the introduction paragraph is first providing a historical view that even qualitative and pre-1900 views encompassed perception as an unconscious inference and as something that was being generated subliminally and updated and then kind of presented to awareness and then the authors place the bayesian brain hypothesis as a kind of formal realization of some of those models of perception and action and explain how active inference is going to be extending those ideas by doing several things not just considering perception as unconscious inference um but also considering various other cognitive features like action planning learning as bayesian statistical issues in how we model them and then by finding approximations and heuristics

that help us address problems that might otherwise be intractable

um

perception as inference

here the bayesian brain perspective is brought up

and does anyone want to like add anything that they've heard or what the bayesian brain kind of brings up for them

one key feature of the bayesian brain um that they're going to point to

is how it contrasts from a kind of outside in signal processing purely recognition based

model so we can think of that model as being descriptive statistics and also being compatible with frequentist statistics like we're getting kind of a blurry picture out there and then we're going to do um

descriptive statistics frequentism and get some sort of sharpened realized image like the camera filters a little blurry

and then we're just going to post process the image and that's going to result in the final image so it's outside

in

um in contrast and this is not novel to active inference this is being described as part of the bayesian brain hypothesis

perception is not a passive outside in process

it is a constructive inside out process in which sensations are used to confirm or disconfirm hypothesis about how they were generative

generated um

the way that that plays out in bayesian inference

is with kind of a tale of two densities

density um being another name for a statistical distribution is like a density

and the tale of two densities was a paper title by ramstead et al it was an early live stream that we did

and the two densities are like the forward and the reverse model which are used in bayesian statistics

as well as expectation maximization and other algorithms and here's what those two densities are

there are the density of observations that's like the observables in the sensory um states and then

there's another density which are like latent states or hidden states that generate those observables

so um if we were looking at a uh red ball then there'd be like the visual sense states of the

location of the ball and then there'd be like a hidden state of the location of the ball so

thermometer giving us readings of the world and then the temperature in the room which is an unobservable

and so we can take thermometer data and we can use that to fit a temperature hidden state and also from

a given hidden state we could use a generative model or what is sometimes referred to as a forward model

to generate predictions about observations in contrast with descriptive statistics which is

taking in observables and kind of distilling them and then saying well the mean was 11 and the variance was

4. if that's the end of the road for that modeler that's the end of the road but the bayesian approach

says okay we've got a mean and a variance and now we're going to be able to generate data

with mean 11 variance 4. so that's the tale of two densities is the way that these two um generative

processes and a recognition model are linked to operationalize that a little bit more the state

of the hidden variables a priori meaning before is called a prior the prior is then going to be updated

by observations and depending on a few factors which we're going to go into you can imagine one extreme

case where whatever observation comes in you just instantly update your prior to that so then it becomes a

posterior like after the fact but the posterior of that first moment it becomes the prior for the

next moment so it's kind of this unfolding process and again one extreme case would be whatever comes

in from the world you just instantly update your prior to that so that's like a zero memory just

instantly updating to that incoming data point like what's the average height in this classroom well

this child was four feet tall okay the average is four now there's five feet okay now it's five feet tall

the other extreme would be like very much denoising the observations by having a really stable maybe even

stubborn hidden state estimate this relates also to attention because when we pay attention to

observables we update our priors more when we're not paying attention observables coming in don't update

our hidden state our hidden state estimates bay's rule tells us how to combine different elements the

prior and the likelihood such that a prior probability becomes transformed into a posterior probability so you have some a priori estimate of how likely it is to be raining at that moment then you're going to detect wetness on your skin or not and then there's going to be a posterior probability update bayesian inference is a broad topic and so there's many resources in place to learn but it's also a very important thing to know in active inference

box 2.1 covers some of the underlying mechanics of bay's theorem

which is like very simple and tautological yet very powerful in its application

there's an example introduced here that's going to come up several more times in the book

and does anyone want to like explain this example with the frogs and the apples like what is happening here

so the situation is that there's a person holding something in their hand and there's only two

options for what it could be either a frog or an apple they also this is their prior belief there's a 10

chance that it's a frog 0.1 because we scale probability distributions to like a total of one

because something must happen but a one for a statistical distribution is like a hundred percent

so there's a 10 chance of it being a frog and a 90 chance of what they're holding being an apple a priori

that's their prior

then they have some sort of likelihood model the likelihood model describes for apples and for frogs

the same time they're holding the same way they're holding the same way they're holding the same

so apples jump one percent of the time that's the probability of action equals jumping vertical line means

conditioned upon so the probability of jumping conditioned upon the hidden state being an apple

is one percent the probability of no jump conditioned on an apple being the hidden state is 99 so apples rarely jump

in contrast the probability of jumping observing a jump when the hidden state is a frog is 81 and 19 for not jumping

so a priori there's a 10 belief that it's a frog 90 that it's an apple that's representing the uncertainty

around what the object is then a jump is observed just intuitively even without any equations if we know

that frogs are more likely to jump than apples observing a jump should increase our favor of that being like a

jumping object and so that's exactly what happens and that's what the bayesian statistics describe

is a situation where you have a prior belief about hidden states that are not directly observable

and you have a likelihood model of how observables which is here why how observables map to different

hidden states and then through the observations you update and realize a posterior belief about hidden states

does anyone have any thoughts or questions about this

we're going to continue on the low road this is a method called exact base so this is kind of like a plug

and chug method where you can take what is described in box one and apply it to the values that are presented here

and you get the exact calculation of different variables depending on what you're trying to find

you're going to be able to calculate basically every variable every unknown you can kind of isolate for and solve

um this is called exact base exact base has um a clear intuition for small state spaces so when it's like

there's two options and there's two for hidden state and two options for behavior it's quite easy to calculate for large state spaces it becomes or unknown state spaces it can become intractable so that's going to motivate some of the heuristics that are brought in later um okay table again people can just please raise your hand if you have any thoughts or questions or want to like dwell at it at a part table 2.1 is going to be uh brock yes uh i just wanted to if anyone is feeling like um so blue is she said there's a lot of math in this uh chapter and um

she's one of the co-ordinators so back to the prince institute and i'm um this is my second time through so um um if you're feeling like this is a lot of math you're not alone but that's okay um

i i just i don't know just wanted to say that like it's it i think the way this is presented here is really helpful for people that have a grasp on some of the other um even simpler parts like the way that you uh made this distinction between the generative model being a thing that's producing predictions and that like that's a really important thing to understand um the bayesian brain versus this sort of bottom-up um um predictive sort of machine thing um

um part of what is happening there though that i guess we're gonna get into a little bit more when he's talking about hidden state spaces is that um like what what that all depends on when he's going through this apple and frog example is that there's um a conditional probability here that like um the reason that you believe that the frog is jumping or whatever or not or might jump is because it's a frog and that may seem like well obviously or like so what like that you know but if it's something a little uh less you know i don't know contrived or something like um what is brock gonna say next or whatever like then it helps maybe to have more priors on what what i'm you know and what i generally you know so i i guess what i'm saying i'm trying to draw some attention to this conditional probability it's um kind of uh like a simple concept but it's kind of what makes that whole thing work at all is that there are probabilities that are kind of like strongly related and it may be part of a hidden state or whatever um but that's kind of what is what is if there weren't any conditional probabilities none of this would work um and that's kind of the uh the magic simple secret sauce there that's that's it's getting traction on so that's helpful or more confusing but thanks for the comments uh the personal comments brock and also yes the conditional probability this vertical line does so much what's the probability that the giants win against the yankees well in my view it's 50 now conditional upon it raining maybe that doesn't matter in your generative model conditional upon this what if i told you that it was seven nothing in the third inning like all these situations where information comes in and there's information that doesn't update our prior like if you uh give a piece of irrelevant information then it doesn't update our prior so like conditioned upon the irrelevant information then we're not updating our hidden state estimate whereas there's other information that comes in that is relevant and that's going to have to do with the sparsity of how the variables are connected and then those conditional estimates help us reduce our uncertainty what's the overall probability that somebody has this has a medical degree in the world it's just the number of medical degrees in the world divided by the total population of the world conditioned upon being in this situation what is it and so that can be a different number

um table 2.1 gives a little bit of a statistical detail on several different forms of distributions now again even for people who have taken several courses or even years of statistics these are not trivial concepts the piece though that we'll just highlight is what a support means and what surprise means support is describing where that distribution is even valid or definable so there's some distributions like the gaussian that have support over all real numbers that's what the fancy $\mathbb{R}$ means so like a gaussian the kind of archetypal case is it's a bell curve centered at zero and it goes off to infinity in both directions with vanishingly small numbers but it has support everywhere so that's the gaussian um in contrast there's other distribution families that for example can only exist on the range between like zero and infinity so that would be like waiting time between events it can't be a negative number so the gaussian is like an inappropriate distribution family to even describe situations that can't be negative surprise is a formula that for any given observation helps you determine how surprising that data point is so we have the children in the classroom and the average height is four feet plus or minus one foot and we observe a new measurement and it's exactly four feet that is like the least surprising observation possible and then as it gets further away from four like the standard deviation increases it's more surprising and so these formula describe for each distribution family how to calculate surprise so that's just the surprise how surprising a new data point coming in is observation given the observer given the hidden states as they're estimated

um okay so that's one notion of surprise that's kind of like new data point comes in how surprising is it there's a second notion of surprise that's called bayesian surprise bayesian surprise is how much your prior moves

following an observation so you could imagine again we have a four foot estimate for the average classroom and data points are coming in and we're surprised or not that's regular surprise

bayesian surprise is the difference between the prior and the posterior so we could get seven foot measurements again and again and again and again but because we're super stubborn in our prior we're still keeping that four foot estimate so those data points are surprising surprising surprising surprising

but the bayesian surprise is zero because the prior is not being updated into a new or different posterior

um that's a little bit on the difference between vanilla surprise naked surprise and bayesian surprise this is

specifically within the bayesian statistics updating framework how do we uh how do we quantify surprise

well we have these formulas you just plug in like the observation and some of the parameters of the

the distribution the parameters are like the values that make the distribution the way it is so for a

gaussian there's two parameters like mean and variance where's the mean of the bell curve the average and how

wide is the bell curve those are parameters surprise can be calculated from the parameters of the distribution and the observation with these equations how do we calculate bayesian surprise

this is going to introduce uh an operation that may be familiar to some or not but it's it's all good

it's called the kl divergence kolbach liber divergence and this is what it looks like

again these are topics where one can go very very deep and learn a lot so just treat it like many coats of paint you're going to know it's a kl divergence because it's D_{kl} just like if it was like f of x is like a function of x

the kl brackets means that we're going to be talking about a kl divergence about what's in the square brackets so at this point i'm going to go over the equations i'm going to find equation 2.3 so we can see it here i'm also going to open up the description so this is really helpful this is a super important and valuable contribution for people who can to create natural language descriptions of equations because it helps get us all on the same page because otherwise this is like inscrutable again even for people who might know the area because the notation can be slightly different with different letters like what is q and so on um so the kl divergence is going to be hinging upon these two vertical lines when it's one vertical line it's going to be used in a lot of different statistic situations to mean conditional upon x x line y x conditioned on y

the two vertical lines are going to be what the kl divergence is being calculated between so this is describing the kl divergence between the q and the p distribution here's q of x this is a function here's p of x that's a function um specifically for bayesian surprise we're talking about the kl divergence between the prior and the posterior so the kl divergence between the prior probability distribution q of x and the posterior probability distribution p of x and then the second part here what does that mean equal sign with a triangle is by definition that's what the triangle means on top is the expectation that's fancy e means expectation over so it's the average value of q of x

of the natural log of q minus natural log of p this part's a little bit less important a little bit more detail but this is where we're going to see it mostly written out as with two vertical lines and the kl divergence is measuring the difference between them if they're the same distribution the difference is zero if they're very different the value is going to be high it's kind of like how much earth do you have to move to reshape the distributions to look like each other it's not exactly that but that's kind of what a kl divergence is so here we can think about the bayesian surprise of that frog jumping brock yes

um i was just gonna say i guess this those details i think this natural log q of x minus which a lot of people that um another way to say that is that that's information those are uh gnats or like a form of bits kind of um so that's the expected difference of information is there zero information or is there some change in other words between the the the p of x being the um prior and the q of x being uh so that divergence when he's about to go into is is that's what we're talking about is like how much space is there informationally between those two things great so a gnat is a unit of measurement for information a bit is a binary bit that's zero one digital computers use bits so people are pretty familiar sometimes with a bit a gnat is a unit that's not based around zero or one it's based around the natural log and base e instead of like a base to binary system so it's it's monotonically related to bits which is to say like they're both going to go up and they're both going to go down but they're not exactly comparable it's like inches and feet um here when we were calculating just regular surprise so how surprising the observation is that could be calculated in nats was it a surprising observation or not did it did it surprise me at all that that can be calculated about surprise here nats are also used to calculate the informational divergence

between the prior and the posterior when we're thinking about bayesian surprise

box 2.2 introduces the fancy E notation fancy E is standing for expectation now sometimes conversationally expectation is talking about the future like what do you expect the weather to be tomorrow in the statistical sense expectation is the average expected value it doesn't necessarily refer to the future it could be an expectation about a future time point but also alone it is only describing basically the average of a distribution so the expectation of a gaussian is the center of the bell curve not like what one believes it's going to be in the future but if someone said what is the expectation for of your in your model for what the average is going to be in 10 time steps that can be addressed so it's compatible with the way that people talk about what do you expect to happen tomorrow but just be aware that expectation alone does not refer necessarily to a future time point it's just referring to a statistical average

and um one note again may be familiar to some maybe the first time hearing for some but it's going to come into play in this book is there are discrete state spaces and continuous state spaces a discrete state space is one where values can only take certain like clickable values like zero one two three four or it could be point one point two point three point four but it's like there are discrete values on a grid that can be achieved like the two options for what the thing in the hand was it's a discrete state space whether it's two or two million it's a finite set of definables and there's no in-betweens in contrast continuous state spaces would be like a continuous variable describing the temperature now one could also describe temperature with a discrete state space in my model i'm describing temperature as either 20 21 22 23 only integer values are allowed for my model so there's advantages and there's challenges for both discrete and continuous state spaces and this book is going to be like juxtaposing them and the chapters even later are going to be like kind of alternating between them because there's active inference in discrete situations discrete state spaces discrete time and then there's also active inference in continuous state spaces and continuous time okay so that was very low down on the low road nothing of the above was like about active inference or even many of the ideas that often bring people to the table that was kind of like an overview on some bayesian statistical ideas bronwyn from the average average average average

With the conditional thing, like in Melbourne tomorrow there's the grand final, so you've got two teams and you said there's a 50-50 chance, but it depends on what conditions. Those conditions, are they mostly subjective?

I mean, they can be subjective and they can be factual as well.

Yeah, I suppose the question is, what states of the conditions are there?

Can they both be, are they mostly subjective or can you have the fact that frogs jump?

That's what frogs do, so that's what generally happens.

So that's not really my subjective opinion of frogs.

But does my subjective opinion put a condition on it?

Does that make sense?

Yeah, great question.

These models in their construction, like what their overall structure is, and then in their parameterization, like what the values are, are subjective.

They're modeler dependent.

And so we might have different perspectives on the overall likelihood of one team to win.

And then somebody might say, it's going to rain tomorrow or it is raining now.

And then in your model, you had that, that's a, going to update your prior, but maybe it doesn't update my prior.

So it is, even when it's like frustratingly obvious, it is still also subjective and multi-perspectival. Yeah.

Okay, great.

That's what I thought.

And then that leads on to the Bayesian surprise, the two differences between surprise and Bayesian surprise.

So I think that, how Bayesian surprise actually operates in the world, so in relationship to that.

Like I might say Geelong's going to win tomorrow.

And even when they've lost, I still think they've won.

And you can't, you can't convince me of anything other than that.

Is that where my prior doesn't change?

Yeah.

So how does Bayesian surprise itself?

Can you give us some examples of how that might operate in the world, in the real world?

Yeah, it's an awesome question.

And to connect really surprised by itself.

So how surprising a single observation is independent of how it changes your belief.

You have a stereotype about some sort of situation, something surprising happens, but then the Bayesian surprise is zero because it's a recalcitrant belief that the person has.

In contrast, there might be a surprising event that induces a Bayesian surprise because someone's like, whoa, I thought about it this way.

And then I was surprised by this observation and that brought me over here.

And then neither of those are the experience of surprise, but that's an area of a lot of research and discussion, which is like, how did these concepts in statistics, like beliefs, surprises, preferences, how do those map onto our psychology and phenomenology?

Is a preference in active inference?

Like I prefer A over B.

Is it expectation related to what people use every day?

And that's kind of the fun part with the active inference ontology is some of these terms are not used in day-to-day conversation.

And so it's like a blank canvas, but in other situations, these are terms that are used day-to-day, ambiguity, attention, belief.

And so are we using them in a technical sense that's compatible or isomorphic with the way that it's used broadly?

Is it incompatible?

And so on.

That's interesting.

I just had a different perception of Bayesian surprise then when you said surprise is always surprise, but Bayesian surprise maps onto, I think you said something like the psychology or the preferences or whatever.

So it's not really surprise.

It's how we manage surprise.

Could it be like that?

What sort of precisions we have on things so that we recognize something or we don't recognize it?

Is it something like that?

Excellent.

And the connection to precision is also technically correct, which is that when observations are coming in and we're treating them precisely, we do have Bayesian surprise because we are updating our priors.

When we have imprecision about observations, we do not update our distributions of beliefs like, oh, that's just a bad thermometer.

I know it says that it's, I believe it's 37 in the room.

The thermometer is just weird.

It says 65, but don't worry about it.

So then that's a surprising observation under the belief that it's 37 plus or minus one, but it doesn't induce a Bayesian surprise because we haven't updated our prior.

Prior.

Yeah.

Okay.

That's interesting.

Yep.

Awesome.

So just to kind of, yeah, continue, please.

It's also interesting in that I watched that, the podcast thing with John Veracquerie the other day.

And it was interesting out of there that just the language, the use of language, and then you were just talking about the ontology and different uses.

And are we actually considering the words that we're hearing as they actually are?

I think that's really, yeah, it's really interesting area with all of this because, and it makes it really, what I've noticed in my, in what I listen to and my studies is that it's like you have to keep going over and over and over until someone says something like you just said then.

Okay.

Okay.

Okay.

That's a weirdo.

That sounds weirdo.

yeah
just to kind of
interesting
oh
it is interesting
so we pay attention to it
we update our priors
we learn
and we deal with the surprising observations
and we deal with the Bayesian surprise
to adjust our regime of attention
and action
along all these ways
and
we merge and connect
with the act in ontology
rather than
spiraling off into
uncertainties that aren't relevant
but that's not to banish uncertainty
like from the realm
it's to have the right course grainings
that help us deal with
certain features of action and perception
here
here's the subjective questions coming up
Bayesian inference is optimal
and this is also the root of like a million discussions
how can it be optimal to crash a car
how can it be optimal to be wrong
I mean we were talking about
how could I be wrong
if betting is always optimal
how could I be wrong
inference is subjective
the results are not necessarily
accurate in any objective sense
Bayesian cognition

and optimal Bayesian calculation
again doesn't mean society accepts that neurodiversity
it means that given that person's generative model
and the observations coming in
or their body
as an embodied generative model
and the observations and constraints coming in
that is its own local optimality
and then there's secondary discussions
around like
is it adequate
is it best amongst a family of counterfactuals
does society accept that
and those are discussions to have
but that's irrational
will not be found
here
and that's a very interesting angle
that kind of reframes a lot of
again rationality
versus irrational thinking
yeah
and this is a second reason
for optimality being thought of as subjective
okay
here we're going to get to figure 2-2
which is a schematic
that's going to be very
different remixes of this schematic
will be found
in many places
here is our model
of temperature in the world
here's the real temperature in the world
hidden state
here's our thermometer
observations coming in
we're updating

and then we're taking actions
which may
over
you know in some way
influence the state of the temperature in the room
so this is like the cybernetic
loop
this is the control theoretic loop
this is describing
a kind of generalized
again not active inference unique way
in which
cognitive entities
take in observations
juxtapose them with the generative model
and then do action selection
to influence the hidden states
such that the observations are brought
into alignment with their preferences
among other things
here's a short discussion
that the state spaces don't have to be the same
between x and x^*
so temperature in the real world
might be like a continuous variable
whereas we might just have a binary state
is it too hot
or not too hot
or you might have a three state
too cold
just right
too hot
or five states
getting too hot
actually
and then you might want to do model selection
across
two

three
four
five
states
internally
and none of that hinges on
quote how it really is
so
there's a lot of degrees of freedom
in generating cognitive models
and their structure
doesn't need to map on to the neuroanatomy of the brain
it doesn't need to map on to a physical mechanism
about the cognitive system
which you may or may not even know
and it doesn't have to be isomorphic
the same structure
as the hidden states of the world
okay
all of this so far
has been purely
inferential
brock
um
i just wanted to
i don't know
leave a bookmark
here
uh
stigmatically
um
you mentioned something about neurodiversity there
in the uh
context of the Bayes optimality
rabbit hole
um
and um
just wanted to remind

that like
the origins
of this come from
uh
psychiatry
and
or the impetus for it
of trying to
um
have a
a causal
reasoning
tool
for addressing
certain kinds of
uh
mental illnesses
so it's
it really
it doesn't
yeah
it just
like you said
it doesn't have to map to
a specific
you know
architecture of brain
arrangement
it's this in the world
or whatever
it's you know
it is very much
like
based on the subjective
model that's already
in there
and um
it

it could
you could
seeing the whole thing
perhaps
like
reason about
how it might make sense
from their perspective
or
many different
kinds of
priors
or whatever
and um
and why
that would be
optimal
for that
model
like
or seem
again
just
yeah
so
great
comments
and to kind
of just
force age
chapter
nine
this is the
meta
bayesian
so it's
the modeler's
subjective

model
the psychiatrist's
subjective
model
of
their
partner's
subjective
model
so
that
is the
genesis of
so many
discussions
around the map
and the territory
and the fallacies
and the fallacies
upon fallacies
and the
cause and effect
and all these
different
discussions
and this is how
this modeling
framework wraps
that up
subjective
model
of a
subjective
model
and that
double
subjectivity
opens up

an unbelievable
amount of
degrees of
freedom in
modeling
which then
refocuses the
imperative
towards selecting
from these
models
but that's a
little bit of a
later point
um
just to
in in the
last minutes
because again
next week
hopefully um
many people
will add
simple
tangential
advanced
formal
any kind
of questions
um
but let's
just in the
last three
or four
minutes
just see
what else
is going

to happen
all of
this up
to the
point
has been
inferential
bayesian
brain
hashtag
live stream
43
they're going
to extend
that inference
about
temperature
so to
speak
into
include
action
how that
happens
is
at the
heart
of
active
inference
in the
next
sections
they're
going to
clarify
the
quantity

that
active
inference
agents
minimize
through
perception
and action
as
variational
free
energy
one note
minimization
and maximization
are like
twins of each
other
because with
just a
negative
sign
they can
be the
same
so
when you
see
minimization
or maximization
you can
just
think
of
finding
relatively
higher
or lower

values
but whether
you choose
to think
of
biologists
talk about
climbing
mountain
probable
like a
Dawkins
book
and a
physicist
might talk
about
finding
the bottom
of an
energy
well
but those
are
going to
be seen
as
not even
just
sisters
or twins
of each
other
but they're
actually
the same
thing
through a

negative
sign
there's
two ways
to reduce
free
energy
as we
will all
come to
learn and
love
you can
update your
model
that's
learning
and perception
and you
can act
in the
world
that's
action
action
and
inference
active
inference
first
you can
learn
this is
a variant
of figure
two
two
where now

we're being
able to see
how the
prediction
and the
observation
are being
put together
there's some
discrepancy
in any
realistic
situation
there will
eventually be
a discrepancy
and then
here are
those two
roads
to reduce
free energy
change
beliefs
which can
be perception
perception
and learning
are much
more
similar
in active
inference
than other
frameworks
like when
the ball
moves across

your visual
field
are you
perceiving
it
or are
you
learning
the
position
of it
as it
moves
over
rapid
time
scales
so
they're
kind
of
the
same
and
can
you
act
here's
how you
can resolve
that
discrepancy
you can
change
your mind
about what
you're
seeing

that's
perception
or
you could
act
that doesn't
mean both
of them are
equally good
or they're
going to
result in
your survival
or they're
plausible
in any
given
situation
so it's
always
situational
what
affordances
there are
for learning
and perception
and for
action
but this
is the
general
framework
how does
free energy
get
quantified
and minimized
we'll return

to this
later
but equation
2.5
and 2.6
are going to
be really
key equations
thankfully
through the
work of
many
people here
we have
really
excellent
natural
language
descriptions
connected
to the
ontology
we also
have a lot
of like
notes
and questions
people have
added
and things
like
derivations
for those
who want
to go
that way
that
Jakob

and others
have
worked
on
so
these
lines
of symbols
can be
read
this way
and with
a mouse
over
we can
see what
those are
so yes
it's like
words
words
words
but
we
have a
staged
development
around
these
equations
because
we're
going to
be
revisiting
them
and
understanding

how
this
F
as the
free
energy
that's
being
minimized
I mean
free
energy
principle
this is
kind of
what
it's
about
how
can
it
be
decomposed
into
these
different
sets
of
natural
language
terms
and
in
what
sense
are
these
terms

being
used
again
we
don't
have
the
minutes
to
describe
but
variational
free
energy
 F
is
about
the
past
and
the
present
equation
 2
 5
and
this
is
what
it
looks
like
a
little
bit
graphically
here's
variational

free
energy
it's
the
same
figure
that
was
2
before
but
now
there's
an
equation
here
when
we
want
to
look
into
the
future
there
are
going
to
be
several
complexities
arising
which
is
that
we're
talking
about

the
surprise
of
observations
we
haven't
observed
yet
and
we're
talking
about
the
consequences
of
actions
that
we
aren't
sure
about
what
their
consequences
will
be
and
so
variational
free
energy
equation
2.5
morphs
into
expected
free
energy

g
equation
2.6
similarly
excellent
work
in
progress
as
everything
is
to
unpack
what
this
is
in
natural
language
there
are
some
special
cases
for
expected
free
energy
such
that
when
certain
terms
are
ignored
or
set
to

zero
we
get
a lot
of
special
cases
that
are
in
quite
broad
use
and
then
that's
the
end
of
the
low
road
equation
2.5
variational
free
energy
past
and
present
equation
2.6
expected
free
energy
g
future
that's

the
end
of
the
low
road
chapter
so
that
has
been
this
first
discussion
on
chapter
2
hope
that
people
have
enjoyed
and
it
would
be
extremely
appreciated
to
write
any
questions
that
you
have
on
chapter
2

so
that
we
can
have
some
specifics
to
follow
up
on
so
good
luck
with
your
learning
in
the
next
week
and
see
you
all
soon
do